

Jack Chang

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EDUCATION

University of California, Irvine

June 2026

Bachelor of Science in Computer Science

Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Computer Vision, ML & Data Mining, Graph Algorithms, ML Optimization, Systems Programming, Software Testing & Quality Assurance

TECHNICAL SKILLS

Languages: C, C++, C#, Python, Java, JavaScript, TypeScript, SQL, Bash, Go, HTML/CSS

Frameworks/Libraries: PyTorch, TensorFlow, scikit-learn, OpenCV, NumPy, Pandas, FastAPI, React, Node.js, Electron.js, REST APIs

Databases, Tools & OS: Supabase, Docker, Kubernetes, CMake, GitHub Actions, Git, Unity, Windows, macOS, Linux

EXPERIENCE

AI Tech Knowledge (applied AI consulting startup)

Jun 2023 – Sep 2024

Part-Time Software Engineer

- Increased throughput for scheduling optimization workloads by **50%+** by developing multithreaded **C++/Python** modules with **Google OR-Tools**
- Saved **20+ minutes per release** by implementing GitHub Actions CI/CD pipelines for automated Windows/macOS builds and installer distribution across **6 GitHub repositories**
- Enabled Windows/macOS deployment by building a cross-platform **Node.js** desktop service to queue, execute, and monitor AI model inference tasks
- Improved reliability of user-server authentication by building an **Electron.js/Supabase** login app with session persistence, logout handling, and secure token handoff to the background solver service

MetaGuru (enterprise digital transformation provider)

Jun 2023 – Nov 2023

Research Scientist Intern

- Reduced repetitive HR policy lookup by **15%** by developing a GPT-4 powered internal Q&A assistant for company policy and employee support
- Evaluated **3 LLM deployment approaches** across prompt engineering, fine-tuning feasibility, and multilingual behavior, ultimately prioritizing a **prompt/RAG-based prototype** for faster iteration and lower maintenance cost

NexStream Technical Education (STEM education startup)

Jan 2022 – May 2023

Intern Data Scientist

- Created the company's first learning-visibility system by implementing a **JavaScript xAPI** event pipeline with Watershed LRS, capturing **20+ product activity events** to help instructors understand student engagement across course offerings
- Reduced manual course review time by **3+ hours weekly** by developing **Python** analysis tools for student progress and open-ended response accuracy

CS Curriculum Developer & Instructor

- Expanded student enrollment by **10%+** by developing and teaching **5 courses** covering Pygame and USACO algorithms
- Improved beginner learning outcomes by designing **20+ project-based lessons** that broke programming and algorithm concepts into interactive exercises

Good Trouble (accessible games technology startup)

Oct 2022 – Mar 2023

Gameplay Programmer Intern

- Reduced projectile performance overhead by **50%** by using object pooling for reusable weapon systems across PC and Xbox-focused gameplay in an unreleased combat racing game using **Unity + C#**

PROJECTS

AI Chess Commentary Generator (project lead)

Feb 2026 – Apr 2026

- Generated **3.9M+** synthetic chess commentaries from **197K+** FEN positions by extracting **27 engine-derived chess features** to translate engine signals into grounded move explanations
- Fine-tuned a T5-small seq2seq model and evaluated it against GPT-4o-mini on **1,000 positions**, earning **69.6%** Gemini-2.5 Flash LLM-judge preference and **73.7%** human preference from chess players

Orion: AI Threat Assessment System (sole developer)

Apr 2026 – May 2026

- Created an AI decision-support simulator that combines a deterministic C++ scoring engine, Python ML ensemble, and LLM/RAG explanations to assess **9 threat scenarios** across object risk, confidence level, and action/no-action tradeoffs